SIMATS ENGINEERING
ASSESSMENT TOOL 1 - Low (Pseudo-code Writing)

COURSE CODE: CSA0903 **COURSE NAME: Programming in Java**

COURSE OUTCOME COVERED

CO1: *Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)*

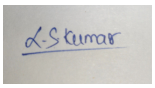
QUESTIONS: 10 **Total Marks: 100**

ANNEXURE A

PSEUDO-CODE WRITING

Code of Conduct:

I L. Sanjay Kumar*(*192425226*) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: 

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CSA0903

1) Student management System

```
START
class student
    PRIVATE name, rollNo, marks
    METHODS acceptDetails()
        INPUT name, rollNo, marks
    END METHOD
    METHOD CalAvg()
        average = marks
    END METHOD
    METHOD display()
        IF average >= 50
            DISPLAY "Pass"
        ELSE
            DISPLAY "Fail"
        END METHOD
    END CLASS
STOP
```

2) Library Book Management

```
START
CLASS Book
    PRIVATE title, author, ISDN, isAvailable
    METHOD issueBook()
        isAvailable = false
    END METHOD
    METHOD returnBook()
        is Available = true
    END METHOD
```

```
METHOD displayDetails()
        DISPLAY title, author, ISBN, isAvailable
END METHOD
END CLASS
STOP
```

Employee Salary Calculation:-

```
START
CLASS Emp
    PRIVATE empID, name, basisPay, designation
    METHOD CalSalary()
            gross = basisPay + HRA + DA
    END METHOD
    METHOD CalNetSalary()
            net = gross - PF - Tax
    END METHOD
    METHOD displayPayslip()
            DISPLAY empID, name, net
    END METHOD
END CLASS
STOP
```

Shape Area Calculation using Polymorphism

```
START
    CLASS Shape
        METHOD CalAreaC)
        END METHOD
    END CLASS
CLASS Circle EXTENDS Shape
        OVERRIDE calAreaC)
END CLASS

CLASS Rectangle EXTENDS Shape
        OVERRIDE CalculateAreaC)
END CLASS
```

```
CLASS Triangle EXTENDS Shape
        OVERRIDE Calarea()
END CLASS
STOP
```

5) Vehicle Registration System

```
START
CLASS Vehicle
        regNo, owner, fueltype
        METHOD displayDetails()
        END METHOD
END CLASS
CLASS Twowheeler EXTENDS Vehicle
        OVERRIDE displayDetails()
END CLASS
CLASS Fourwheer EXTENDS Vehicle
        OVERRIDE displayDetails()
END CLASS
STOP
```

6) Bank Account Hierarchy

```
START
CLASS BankAcc
        accNo, holder, balance
        METHOD Withdraw()
        END METHOD
ENDCLASS
CLASS SaveAcc EXTENDS BankAcc
        OVERRIDE Withdraw()
END CLASS
CLASS CurrentAcc EXTENDS BankAcc
        OVERRIDE Withdraw()
ENDCLASS
STOP
```

Hospital Patient Record System.

```
START
Class Patient
    PRIVATE diagnosis, prescription
    METHOD generateBILL()
    END METHOD
END CLASS
CLASS InPatient EXTENDS Patient
    OVERRIDE generateBILL()
END CLASS
CLASS OutPatient EXTENDS Patient
    OVERRIDE generateBill()
END CLASS
STOP
```

E-Commerce Product Catalog

```
START
class Product
    Method applyDiscount()
    END METHOD
END CLASS
CLASS Electronics EXTENDS Product
    OVERRIDE applyDiscount()
END CLASS
CLASS Clothing EXTENDS Product
    OVERRIDE applyDiscount()
END CLASS
CLASS Grocery EXTENDS Product
    OVERRIDE applyDiscount()
END CLASS
STOP
```

9) University staff Hierarchy

```
START
class staff
        PRIVATE staffID, name
        PROTECTED salary
        METHOD calall()
        END METHOD
END CLASS
CLASS Teach EXTENDS staff
        OVERRIDE calall()
END CLASS
CLASS lecture EXTENDS Teach
        OVERRIDE calall()
END CLASS
STOP
```

10) Smart Home Device Controll

```
START
class Smarthome
        Method executeCommand(cmd)
        END Method
END CLASS
class light EXTENDS Smarthome
        OVERRIDE executeCommand(cmd)
END CLASS
CLASS Thermostat EXTENDS Smarthome
        OVERRIDE executeCommand(cmd)
END CLASS
STOP
```

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient